Hand in 2

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Sorry for the miniuscule text, the tables overflowed when I tried to make it bigger. Also, the links to propositions and such do not provide much contrast to the cell colors, but they should be clickable in most PDF readers.

Exercise 1(d–j): Properties of particular spaces

	I_o^2	$\mathbb{R}_\ell$	$\mathbb{R}_\ell^2$	$\mathbb{R}^\omega$ product	$\mathbb{R}^\omega$ uniform	$\mathbb{R}^\omega$ box	$\mathbb{R}^{[0,1]}$ product
Connected	Example 6, §24	$[0,\infty)$	$[0,\infty)\times\mathbb{R}_\ell$	Exercise 10, §23	See proposition 1.1	Example 6, §23	Exercise 10, §23
Path connected	Example 6, §24	↑	↑	Interpolation	Implied by above	↑	
Locally connected	Exercise 3, §25	$[y,\infty)$	$\prod_i[y_i,\infty)$	↓	See proposition 1.2	*Implied by below*	
Locally path connected	Exercise 3, §25	↑	↑	Interpolation	*See proposition 1.2*	*See proposition 1.7*	
Compact	Thm 27.1	$\{[n,n+1)\}_{n\in\mathbb{Z}}$	←	π_i onto $\mathbb{R}$	π_i onto $\mathbb{R}$	π_i onto $\mathbb{R}$	π_i onto $\mathbb{R}$
Locally compact Hausdorff	Example 3, §29	*See proposition 1.5*	*See corollary 1.1*	Example 2, §29	Exercise 4, §29	*Example 2 §29*	Example 2, §29 (adapt)
Hausdorff	Order topology	$x\neq y\to[x,t),[t,y+1)$	Thm 31.2(a)	Thm 31.2(a)	Metrizable	Refinement of product	Thm 31.2(a)
Regular	↓	↓	↓	↓	↓		Thm 31.2(b)
Completely regular	↓	↓	Example 1, §33	↓	↓		Thm 33.2
Normal	Thm 32.3	Example 2, §31	Example 3, §31	Thm 32.2	Thm 32.2	N/A	Example 1, §32
Metrizable	See proposition 1.3	*see Exercise 5(a), §30*	←	Thm 20.5	Uniform metric		
Second-countable	See proposition 1.4	*See proposition 1.6*	←	Thm 30.2	Example 2, §30	Example 2, §30 (adapt)	$\{0,1\}^{[0,1]}$?

Exercises 2–6: General preservation properties

	Metric space	Compact Hausdorff	Arbitrary subspace	Open subspace	Closed subspace	Finite products	Countable products	Arbitrary products	Continuous image
Connected	$\{0,1\}_{\text{disc}}$	$\{0,1\}_{\text{disc}}$	$X\cup Y$	$X\cup Y$	$\{0,1\}\subset[0,1]$	Thm 23.6	Exercise 10, §23	Exercise 10, §23	Thm 23.5
Path connected	↑	↑	$\mathbb{Q}\subseteq\mathbb{R}$	$U\cap V=\varnothing\Rightarrow U\cup V$ not connected	Topologists sine curve	→	→	See proposition 1.8	$f\circ p$
Locally connected	$\mathbb{Q}$	Topologists sine curve	$\mathbb{Q}\subseteq\mathbb{R}$	↓	Topologists sine curve	↓	See proposition 1.9	←	$\mathbb{R}$ discrete → $\mathbb{R}$ canonical
Locally path connected	↑	↑	$\mathbb{Q}\subseteq\mathbb{R}$	See proposition 1.11	↑	See proposition 1.9	↑	↑	↑
Compact	$\mathbb{R}$	By hypothesis	$(0,1)\subset[0,1]$	$(0,1)\subset[0,1]$	Thm 26.2	Tychonoff	Tychonoff	Tychonoff	Thm 26.5
Locally compact Hausdorff	↑	Compact → locally compact				See proposition 1.10	$\mathbb{R}^\omega$ not locally compact	←	↓
Hausdorff	Metric	By hypothesis	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	$\{0,1\}$ discrete → $\{0,1\}$ indiscrete
Regular	↓	↓	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	See example 1.1
Completely regular	↓	↓	Thm 33.2	Thm 33.2	Thm 33.2	Thm 33.2	Thm 33.2	Thm 33.2	↑
Normal	Thm 32.2	Thm 32.3	→	See example 1.2	Closed in subspace ⇒ closed in ambient	$\mathbb{R}_\ell^2$	←	Example 1, §32	Discrete space → non-normal space
Second-countable	$\mathbb{R}^\omega$ uniform	Any uncountable discrete space	Thm 30.2	Thm 30.2	Thm 30.2			$\mathbb{R}^{[0,1]}$	$\mathbb{R}\to\mathbb{R}/\mathbb{Z}$ quotient
Metrizable	By definition	Urysohn Metrization Thm.	Restriction of metric	Restriction of metric	Restriction of metric			$\mathbb{R}^{\mathbb{R}}$ not first countable	See example 1.1

1 Proofs

Proposition 1.1. *$\mathbb{R}^\omega$ with the uniform topology is **not connected**.*

Proof. The countably infinite product of the real line $\mathbb{R}^\omega$ with the uniform topology is the space induced by the uniform metric

$$\bar{d}(x, y) = \sup_{n \in \mathbb{N}} \min(|x_n - y_n|, 1)$$

where $x = (x_n)$ and $y = (y_n)$ are sequences of real numbers.

We claim this space is **not connected**.

Let B be the set of all bounded sequences, and let U be the set of all unbounded sequences,

1. $B = \{x \in \mathbb{R}^\omega \mid \exists M > 0 \text{ s.t. } |x_n| \leq M \ \forall n \in \mathbb{N}\}$
2. $U = \mathbb{R}^\omega \setminus B$.

Clearly, these are nonempty since

$$(0, 0, 0, \ldots) \in B$$

while

$$(1, 2, 3, \ldots) \in U.$$

B is open.

Let $x = (x_n) \in B$. Then there exists some $M_x > 0$ such that $|x_n| \leq M_x$ for all $n \in \mathbb{N}$.

Consider the ball of radius 1 around x

$$B_{\bar{d}}(x, 1) = \{y \in \mathbb{R}^\omega \mid \bar{d}(x, y) < 1\}.$$

If $y \in B_{\bar{d}}(x, 1)$, then

$$\sup_{n \in \mathbb{N}} \min(|x_n - y_n|, 1) < 1$$

implying that for every n, $|x_n - y_n| < 1$ for all n. Using the triangle inequality we can attain the following bound:

$$|y_n| \leq |y_n - x_n| + |x_n| < 1 + M_x.$$

Thus the sequence y is bounded, meaning

$$B_{\bar{d}}(x, 1) \subseteq B.$$

Since every point of B has an open neighborhood completely contained in B, B is open.

U is open.

Let (x_n) be undounded. Consider the same open ball $B_{\overline{d}}(x, 1)$. If $y \in B_{\overline{d}}(x, 1)$ then

$$|y_n| \geq |x_n| - |x_n - y_n| > |x_n| - 1$$

so y is unbounded since $|x_n|$ can be arbitrarily large.

Therefore, $B_{\overline{d}}(x, 1) \subseteq U$ is open.

Then

$$\mathbb{R}^\omega = B \cup U.$$

■

Proposition 1.2. *$\mathbb{R}^\omega$ with the uniform topology is **locally connected**. In fact, it is **locally path connected**.*

Proof. We claim the space is **locally path connected**.

For any point $x = (x_n) \in \mathbb{R}^\omega$ and any radius $\varepsilon > 0$ the open ε-ball centered at x is

$$B_{\overline{d}}(x, \varepsilon) = \{y \in \mathbb{R}^\omega \mid \overline{d}(x, y) < \varepsilon\}.$$

If we choose $\varepsilon < 1$, the uniform metric forces every coordinate to be within ε of x_n. Therefore, the open ε-ball is

$$B_{\overline{d}}(x, \varepsilon) = \prod_{n \in \mathbb{N}} (x_n - \varepsilon, x_n + \varepsilon).$$

Take any point $y \in B_{\overline{d}}(x, \varepsilon)$. Construct a linear path $f : [0, 1] \to B_{\overline{d}}(x, \varepsilon)$ from the center point x to y by setting

$$f(t) = (1 - t)x + ty.$$

Since each interval $(x_n - \varepsilon, x_n + \varepsilon)$ is convex, every point remains inside the ball for all $t \in [0, 1]$.

Since every point has a neighborhood basis consisting of these path-connected open balls, the space is locally path-connected which makes it locally connected.

■

Proposition 1.3. *The ordered square I_o^2 is **not metrizable**.*

Proof sketch. We invoke the following theorem:

Theorem. *If a topological space is **compact** and **metrizable**, then it must be separable.*

So it will suffice to elucidate that the space is compact and not separable.

The Heine-Borel theorem for ordered spaces says that a linearly ordered topological space is compact if and only if it has a least element, a greatest element, and satisfies the least upper bound properrty.

The ordered square has minimum $(0,0)$, maximum $(1,1)$. Because the standard unit interval has the least upper bound property, the lexicographic order preserves this completeness. Thus I_o^2 is compact.

The space is not separable. We will find an uncountable collection of open sets that are completely isolated from each other. For every real $x \in (0,1)$ consider the vertical segment at x, excluding endpoints,

$$V_x = \{x\} \times (0,1).$$

In the dictionary order, V_x is the open interval $((x,0),(x,1))$. Because it is an open interval, V_x is an open set in the dictionary order topology. Notice that if $x \neq y$ the vertical segments V_x and V_y are disjoint,

$$V_x \cap V_y = \emptyset.$$

Since there are uncountably many choices of $x \in (0,1)$ we have an uncountable collection of pairwise disjoint non-empty open sets.

Suppose then that some dense $D \subseteq I_o^2$ exists. By definition the set must intersect all non-empty open sets, but there are uncountably many so this cannot be the case.

■

Proposition 1.4. *The ordered square I_o^2 is **not second-countable**.*

Proof sketch. As we saw in the proof of proposition 1.3, we can construct an uncountable collection of pairwise disjoint, non-empty open sets. By definition, each of these sets must contain at least one basic open set from the basis.

Since the vertical intervals are disjoint from one another, a basic open set in one cannot be shared by any other, meaning we must have uncountably many basis elements. ■

Proposition 1.5. *The Sorgenfrey line $\mathbb{R}_\ell$ is **not locally compact Hausdorff**.*

Proof by some guy on mathstackexchange said. In this Math stackexchange post there is given a proof that the Sorgenfrey line is not locally compact, and thus not locally compact Hausdorff. ■

Corollary 1.1. *The Sorgenfrey plane $\mathbb{R}_\ell^2$ is **not locally compact Hausdorff**.*

Proof sketch. Suppose it was. Considert the standard projection map onto coordinate one,

$$\pi_1 : \mathbb{R}_\ell \times \mathbb{R}_\ell \to \mathbb{R}_\ell$$

where

$$\pi_1(x, y) = x.$$

This map is continuous and open. The image of a locally compact space under a continuous open map is also locally compact, meaning $\mathbb{R}_\ell$ is locally compact, contradiction proposition 1.5.

Thus $\mathbb{R}_\ell^2$ is not locally compact and thus not locally compact Hausdorff. ■

Proposition 1.6. *The Sorgenfrey line $\mathbb{R}_\ell$ is **not second countable**.*

Proof. Suppose it is second-countable with basis

$$\mathscr{B} = \{B_1, B_2, \ldots\}.$$

For every $x \in \mathbb{R}$, $[x, x+1)$ is open. Since $\mathscr{B}$ is a basis there must exist, for every real $x \in \mathbb{R}$, an element $B_{n_x} \in \mathscr{B}$ such that

$$x \in B_{n_x} \subseteq [x, x+1].$$

Notice that x must be the absolute least element in $[x, x+1)$. Since $B_{n_x} \subseteq [x, x+1]$ and $x \in B_{n_x}$, x must be the absolute least element of B_{n_x}.

If we have two distinct $x, y \in \mathbb{R}$, then they cannot share the same basis element. If $B_{n_x} = B_{n_y}$ then these basis elements have two distinct minimum values, which is impossible.

Thus we can construct an injection $x \mapsto B_{n_x}$, meaning

$$\aleph_1 = |\mathbb{R}_\ell| \leq |\mathscr{B}|$$

so $\mathscr{B}$ is uncountable. ■

Proposition 1.7. *$\mathbb{R}^\omega$ with the box topology is **not locally path connected**.*

Proof. Let U be a basic open neighborhood of 0 by

$$U = \prod_{n \in \mathbb{N}} (-1/n, 1/n).$$

Pick a point $x = (x_1, x_2, \ldots) \in \mathbb{R}^\omega$ by setting

$$x_n = \frac{1}{2n}.$$

Since $x_n \in (-1/n, 1/n)$ for all $n \in \mathbb{N}$ we have $x \in U$.

Suppose a path $f : [0, 1] \to U$ with $f(0) = 0, f(1) = x$ exists.

Then

$$f_n = \pi_n \circ f : [0, 1] \to (-1/n, 1/n)$$

is a continuous real valued function with $f_n(0) = 0$ and $f_n(1) = 1/2n$.

By the intermediate value theorem, each f_n attains all values between 0 and $1/2n$.

In particular, there exists $t_n \in [0, 1]$ so

$$f_n(t_n) = \frac{1}{4n}.$$

We get a sequence $\{t_n\}_n \subseteq [0, 1]$, a compact metric space. Let $t^* \in [0, 1]$ be the limit of some convergent subsequence. Assume without loss of generality that

$$\lim t_n = t^*.$$

Let $f(t^*) = (y_1, y_2, \ldots) \in U$. Take

$$V = \prod_{n \in \mathbb{N}} (y_n - \varepsilon_n, y_n + \varepsilon_n)$$

where

$$\varepsilon_n = \frac{1}{2}\left|\frac{1}{4n} - y_n\right| < |f_n(t_n) - y_n|.$$

If $y_n = 1/4n$ set $\varepsilon_n = 1/8n$.

Then

$$\left|\frac{1}{4n} - y_n\right| \geq \varepsilon_n > 0.$$

Since V is open, there exists $\delta > 0$ so

$$|t - t^*| < \delta \Rightarrow f(t) \in V.$$

Since $\lim t_n = t^*$ choose N large enough so

$$|t_n - t^*| < \delta,\ n \geq N.$$

Then $f(t_n) \in V$.

Now, for all $n \geq N$ we have

$$\pi_n(f(t_n)) \in (y_n - \varepsilon_n, y_n + \varepsilon_n)$$

meaning

$$|f_n(t_n) - y_n| < \varepsilon_n.$$

Substituting we get

$$\left|\frac{1}{4n} - y_n\right| < \varepsilon_n,$$

but this contradicts

$$\left|\frac{1}{4n} - y_n\right| \geq \varepsilon_n.$$

■

Proposition 1.8. *If X_i, ranging over indices $i \in I$ are all path connected. Then*

$$X := \prod_{i \in I} X_i$$

is path connected *(with the product topology).*
Note: This fails quite spectacularly if we assume the box topology.

Proof sketch. Recall that the basis is

$$\mathscr{B} = \left\{ \prod_{i \in I} U_i \;\middle|\; U_i \text{ open in } X_i \text{ and } U_i \neq X_i \text{ for finitely many } i \in I \right\}.$$

Take some $x = (x_i)_{i \in I},\ y = (y_i)_{i \in I} \in X$.

Then, since X_i is path connected for each $i \in I$, we have some continuous $f_i : [0,1] \to X_i$ with

$$f(0) = x_i, \text{ and } f(1) = y_i.$$

Construct the path $f : [0,1] \to X$ by

$$f(t) = (f_i(t))_{i \in I}.$$

This is nice and well-defined because I say so and we have

$$f(0) = (f_i(0))_{i \in I} = (x_i)_{i \in I}$$

and

$$f(1) = (f_i(1))_{i \in I} = (y_i)_{i \in I}.$$

By the universal property of the product topology, f is continuous if and only if $\pi_i \circ f = f_i$ is continuous for all $i \in I$.

Therefore f is continuous. ■

Proposition 1.9. *Finite products of locally connected spaces* ***are locally path connected****, whereas countably infinite products, and thus arbitrary products, of locally connected spaces fail to be even locally connected.*

Proof. Suppose X_i, with $i \in [n] := \{1, 2, \dots, n\}$, are locally path connected.

Set

$$X = \prod_{i \in [n]} X_i.$$

A sub-basis can be written as $\pi_i^{-1}(U_i)$ where $u_i \subseteq X_i$ is open.

A basic open set $W \subset X$ containing some point $x = (x_1, x_2, \ldots)$ can be written as

$$W = \prod_{i \in [n]} U_i$$

where U_i is an open neighborhood of x_i.

Since X_i is locally path connected, for each $x_i \in U_i$ we can find a path connected open neighborhood $V_i \subset X_i$ such that

$$x_i \in V_i \subseteq U_i.$$

Then

$$V = V_1 \times V_2 \times \cdots \times V_n$$

is an open subset of X in the product topology.

It is path connected since, as has been established, the product of path connected spaces is path connected.

Thus X is locally path connected.

Counterexample for countably infinite product.

Take $X = \{0, 1\}^{\mathbb{N}}$, i.e. the Cantor space with the product topology.

Each $X_n = \{0, 1\}$ are trivially locally path connected and thus locally connected.

Let $C \subseteq X$ be connected.

For each $n \in \mathbb{N}$ consider the restriction

$$\pi_n \big|_C : C \to \{0, 1\}.$$

Since π_n is continuous and C is connected, the image must be, but $\{0, 1\}$ is discrete so its onoly nonempty connected subsets are singletons.-

Consequently, for each n, there exists $c_n = 0, 1$ such that

$$\pi_n(C) = \{c_n\}.$$

Thus every element of C has the same coordinates, so

$$C = \{(c_n)_{n \in \mathbb{N}}\}.$$

Thus X is totally disconnected.

■

Example 1.1.
regular $\not\to$ regular & metrizable $\not\to$ metrizable
Let $X = \mathbb{R}$ with the standard topology, a metric space and thus regular. Furthermore the space is metrizable.

Let $Y = \{0, 1\}$ with the trivial topology.
Let $f : \mathbb{R} \to \{0, 1\}$ with

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

We have $f^{-1}(Y) = \mathbb{R}$, open, and $f^{-1}(\emptyset) = \emptyset$, open.
Y is not regular since it is not $T - 1$, $\{0\} = Y \setminus \{1\}$ and $\{1\}$ is not open.
Furthermore, Y is not metrizable since it is not Hausdorff.

Proposition 1.10. *The finite product of **locally compact** spaces is **locally compact**.*

Proof sketch. Let X_i for $i = 1, 2, 3, \dots, n$ be locally compact.

Then every point in each X_i has a compact neighborhood.

Take any point $x = (x_1, x_2, \dots)$. For each $x_i \in X_i$ choose compact neighborhood $K_i \subseteq X_i$.

The cartesian product

$$K = \prod_{i \in [n]} K_i$$

is compact (apply Tychonoff's theorem or go directly from the fact that the product is finite).

Moreover, K is a neighborhood of x in the product topology.

Thus, every point has a compact neighborhood, making the finite product locally compact. ∎

Proposition 1.11. *An open subspace of a locally path connected space is locally path connected.*

Proof sketch. Let X be locally path connected and $Y \subseteq Y$ open.

Let $x \in Y$ and $U \subseteq Y$ and open neighborhood of x. Then, since Y is open, U is open in X. Thus there exists $W_U \subseteq U$ path connected such that

$$x \in W_U \subseteq U,$$

but

$$U \subseteq Y$$

so

$$W_U \subseteq Y$$

is open path connected in Y. ∎

Example 1.2. Let $X = \{a, b, c, d\}$ with $\mathcal{T} = \{\emptyset, X, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}\}$. X is normal as there exist no disjoint nonempty closed sets (they all have a in common). The space is normal, but $Y = \{b, c, d\}$ (open) is not since $\{b\}, \{d\}$ cannot be separated into disjoint open sets,

$$\mathcal{T}_Y \setminus \{\emptyset, Y\} = \{\{d\}, \{b, d\}, \{c, d\}\}.$$